

Airbnb

Understanding Cities Through Data

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Based on an original assignment by Enric Junqué de Fortuny

“We’re a community-driven brand, but at the same time, we want every host in every home to recognize that they’re all individuals, and to use Airbnb as an expression of their individuality.”

— Joe Gebbia, co-founder of Airbnb

Introduction

Airbnb is a global platform that connects travellers with hosts offering short-term stays in private homes. Prices are set by hosts, while Airbnb takes a commission from both sides of each transaction. Since launching in 2008, the company has scaled rapidly, now listing over 7 million properties across more than 220 countries and regions.

Inside Airbnb is a mission-driven initiative that collects, analyses, and shares publicly accessible Airbnb data to help citizens, policymakers, and researchers understand how the platform affects housing and neighbourhoods. The project advocates for transparency and empowers communities to make informed decisions about the role of short-term rentals in residential areas.

The Scenario

You are a newly hired data scientist. Your senior analyst has asked you to conduct a preliminary analysis of the Airbnb market in a city of your choice. They need to be able to open your work, understand what you did and why, and build upon it. This assignment will test your ability to analyze and interpret real estate data in a practical business context.

You can choose either the perspective of Airbnb or that of a third party (government, individual,...) For example, you could help the platform improve host performance or pricing strategy in your chosen city, assess the impact of short-term rentals on housing availability and local neighborhoods, or identify the most promising segments, neighborhoods, or listing types to maximise your individual revenue through the app.

This document was prepared by Tamara van der Does in June 2026, based on an original assignment by Enric Junqué de Fortuny. It is designed to promote class discussion and learning outcomes rather than to illustrate effective or ineffective management of a given situation.

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Deliverables

On blackboard, submit a **zip file** named `Airbnb-[FirstName]-[LastName].zip` by **Wednesday at noon**. It must contain:

1. **Your project folder** including the README, jupyter notebook file, raw data, and outputs.
2. **A one-page memo** (.pdf) formatted as a professional email to your senior analyst, summarizing your findings and recommendations so they can build on it before the client meeting.

Project folder

Set up your folder with a clear structure that the senior analyst will be able to recognize. Figure out the organization, question, expectations, to do's, and potential limitations before starting to code. These might change during your project, which you can describe in the "Things tried" section. **Choose a city you know well** to be able to develop a creative question and hypotheses.

README File

Write a short README.md (10–15 lines). Cover: your city, the business question and chosen perspective; how the folder is organized; what data you used and where it came from (with date downloaded); how to run the notebook (any additional prerequisites installs needed); a high-level description of what your notebook does; and your name and contact.

Jupyter Notebook header

The **first cell** of your notebook must be a markdown cell. Before writing any code, fill in:

Question:	The business problem you are trying to solve
Hypotheses:	Your initial expectations, written before any analysis (specific, measurable, actionable, relevant, time bound)
To Do:	Planned steps (update as you go and leave a few future steps remaining)
Things tried:	What did not work and what you learned
AI used:	Which tools and for what

Data & Exploration

Download the Airbnb dataset for your chosen city from <https://insideairbnb.com/get-the-data/>. Use `listings.csv` as your primary source. You may optionally incorporate `calendar.csv` for time-series analysis (Session 4).

Include the most interesting **exploration findings** in your notebook (shape, key distributions, anything surprising) with a markdown cell after each block explaining what you see and why it matters.

Analysis

Complete the analysis using the tools covered in Sessions 1–4. Include:

1. **Data cleaning:** types fixed, missing values handled with documented decisions, duplicates removed, derived columns explained. Save cleaned data.
2. **Relevant quantitative analysis:** `groupby`, filtering, aggregation, and plots to answer your business question.
3. **Markdown cells** explaining the reasoning at each step (not just what the code does, but why).
4. **Sanity checks:** sample size, recoding, and unusual values verified and visible in output throughout.

Important notes. Use Claude, ChatGPT, or similar tools freely for debugging and plot styling, but you **SHOULD** use the packages covered in class unless you have a strong reason otherwise (justify it), and you **SHOULD NOT** rely on AI for ideation or interpretation. Log what you used it for in the notebook header. Finally, do not use screenshots for code or text.

The Memo

Write a **one-page memo** formatted as a professional email to your senior analyst. Think of it as the message you attach to the zip file: concise, direct, written so I can read it in two minutes and know exactly what you found and what you recommend.

- **Summary of work:** one paragraph covering the perspective, the questions, what you did, and whether your initial assumptions held up.
- **Key insights:** 2-3 data-backed initial findings, each leading with the insight and including the supporting number.
- **Business implications and recommendations:** what these findings mean for the client and what specific actions you recommend.
- **Limitations and next steps:** What you were not able to do and what is missing from the data.

A suggested template is provided at the end of this document.

File name: Airbnb-[FirstName]-[LastName]-memo.pdf

Evaluation Criteria

Total: 40 points. This assignment counts for 40% of your final grade. Breakdown: Workflow 10 · Analysis 15 · Memo 15.

Category	Criteria	Points
Workflow (10 pts)	Clear README file and data structures.	4
	Notebook header reflecting work process.	3
	Notebook restarts cleanly end-to-end with explanations.	3
Analysis (15 pts)	Originality of hypothesis, local validity.	3
	Proper data cleaning and exploratory data analysis.	3
	Sanity checks visible in output.	3
	Effective use of visualizations, tied into the hypothesis.	3
	Interpretation of results.	3
Memo (15 pts)	2–3 data-backed insights.	4
	Business relevance and actionable recommendations.	4
	Honest limitations and thoughtful new hypotheses.	4
	Structure, clarity, tone.	3
Total		40

Example Memo

FROM: *[Your Name]*

SUBJECT: *[One sentence stating the core finding — e.g., “Central listings in Barcelona are underpriced by ~30% relative to comparable markets”]*

1. Summary of Work

[One paragraph. State the perspective, the questions you set out to answer, the main analytical steps, and whether your initial assumptions held up or were contradicted by the data.]

2. Key Insights

Insight 1: *[Lead with the finding, not the chart. E.g., “Entire-home listings near the historic centre are systematically underpriced — median €85/night vs. €120 in comparable European cities.”]*

Insight 2: *[e.g., “Superhosts command an ~18% pricing premium but represent only 12% of listings in the highest-demand neighbourhoods.”]*

3. Business Implications and Recommendations

[What do these findings mean for the client? Connect the numbers to decisions they can actually make.]

Recommendation 1: *[Specific and actionable, tied to your data. E.g., “Launch a dynamic pricing nudge for hosts in [neighbourhood], targeting the €20 gap identified above.”]*

Recommendation 2: *[...]*

4. Limitations and next steps:

[What the data may not capture; key assumptions; what future studies you recommend; other possible hypotheses]